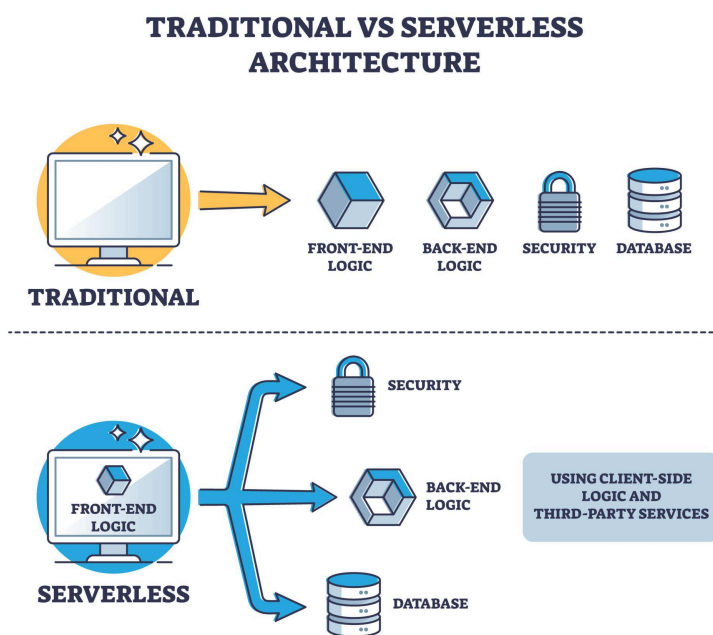


Assignment 3: Trends and Cloud Services Overview

This report discusses modern trends in software architecture and cloud computing. It elaborates on serverless architecture, Progressive Web Apps (PWAs), and the role of Artificial Intelligence (AI) and Machine Learning (ML) in software systems. Additionally, it explains cloud computing service models—SaaS, PaaS, and IaaS—along with their use cases.

1. Serverless Architecture

Serverless architecture is a cloud computing model where developers focus on writing code without managing servers or infrastructure. The cloud provider automatically handles server provisioning, scaling, and maintenance.

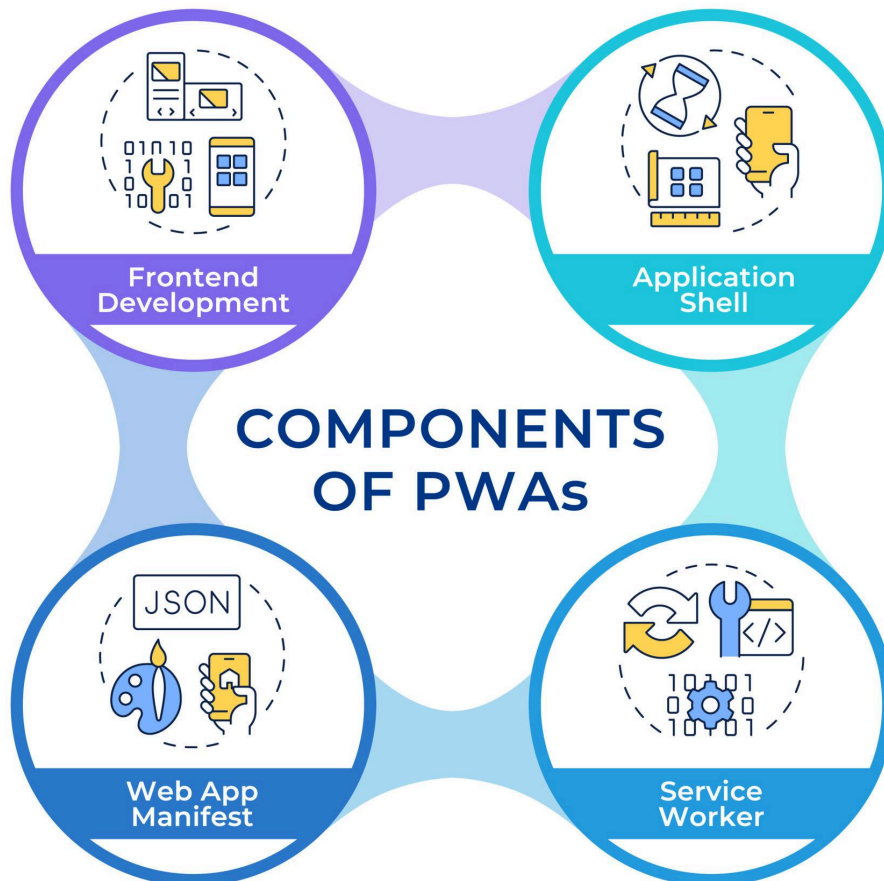


Benefits of Serverless Architecture:

- Reduced operational overhead as infrastructure management is handled by the cloud provider.
- Automatic scaling based on workload demand.
- Cost efficiency since users pay only for actual execution time.
- Faster development and deployment cycles.
- High availability and fault tolerance by design.

2. Progressive Web Apps (PWAs)

Progressive Web Apps combine the best features of web and mobile applications. They are built using standard web technologies but offer native app-like experiences.



Key Characteristics of PWAs:

- Offline functionality using service workers.
- Fast loading and responsive user interfaces.
- Installable on user devices without app stores.
- Secure communication via HTTPS.
- Automatic updates.

PWAs are ideal for applications that require broad accessibility, lower development cost, and consistent performance across devices.

3. Role of AI and Machine Learning in Software Architecture

AI and Machine Learning play a significant role in modern software architecture by enabling systems to learn from data and make intelligent decisions.

Applications of AI and ML in Software Architecture:

- Personalized user experiences through recommendation systems.
- Predictive analytics for business insights.
- Intelligent automation and decision-making.
- Enhanced security through anomaly detection.
- Chatbots and virtual assistants.

AI-driven architectures often use microservices and event-driven designs to handle large-scale data processing and real-time inference.

4. Cloud Computing Service Models

4.1 Software as a Service (SaaS)

SaaS provides fully functional software applications delivered over the internet. Users access the software via browsers without managing infrastructure or platforms.

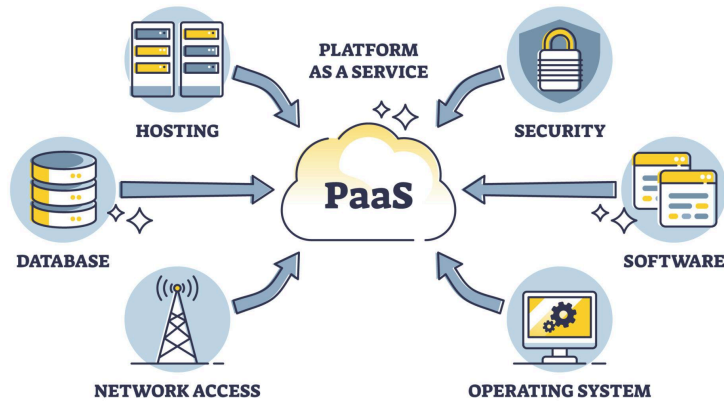


Use Cases:

- Email services (e.g., Gmail).
- Customer relationship management (CRM) systems.
- Collaboration tools like document editors.

4.2 Platform as a Service (PaaS)

PaaS provides a platform that allows developers to build, deploy, and manage applications without worrying about underlying infrastructure.

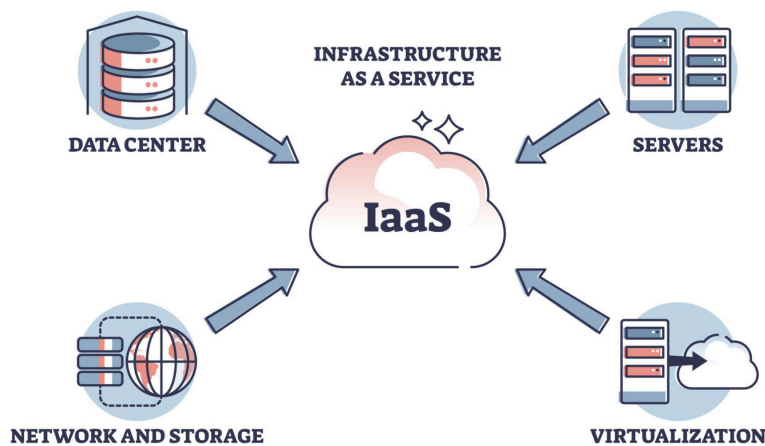


Use Cases:

- Web application development.
- API development and deployment.
- Application testing and CI/CD pipelines.

4.3 Infrastructure as a Service (IaaS)

IaaS provides virtualized computing resources such as servers, storage, and networking. Users have full control over the infrastructure.



Use Cases:

- Hosting enterprise applications.
- Disaster recovery and backup.
- High-performance computing workloads.

Conclusion

Modern software systems increasingly rely on cloud-native architectures, serverless models, and intelligent systems powered by AI and ML. Understanding these trends and cloud service models enables organizations to build scalable, efficient, and future-ready applications.